DBMS Project Task - 4

Project Group Number - 47
Tutorial Group Number - 4
Tutorial Room Number - C210
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Citizen Social Responsibility & Trust Score Management System

User Types/Dashboard Types

- **Citizens -** Represent Normal Users which can see their scores, actions, restrictions, history, benefits, file complaints etc.
- **Admin -** Manages System and its rules, manages users, functionality of the system, definitions(addition/deletion) of different entities etc. (Simple Addition/Deletion/View Operations).
- **Government Official -** Record/Report actions, review complaints, impose restrictions, update scores etc.
- **Analyst -** Study and Analyse things as a whole as per the requirements, study trends, access/generate reports, filter areas/citizens/actions across different parameters.

Note - Values in the below queries can be switched by placeholder “?” when using in backend logic. Values are used to fetch some real data.

Note - Primary Keys are “auto increment” and hence maybe not used in inserting queries listed below.

Citizen Dashboard Queries (10 Queries which a citizen dashboard can use)

01. View Current Score

```sql
SELECT CurrentScore
FROM citizen
WHERE CitizenID = 101;
```

```
mysql> SELECT CurrentScore
    -> FROM citizen
    -> WHERE CitizenID = 101;
+--------------+
| CurrentScore |
+--------------+
|           85 |
+--------------+
1 row in set (0.00 sec)
```

02. View Profile

```sql
SELECT FirstName, LastName, City, ContactNo
FROM citizen
WHERE CitizenID = 101;
```

```
mysql> SELECT FirstName, LastName, City, ContactNo
    -> FROM citizen
    -> WHERE CitizenID = 101;
+-----------+----------+-------+------------+
| FirstName | LastName | City  | ContactNo  |
+-----------+----------+-------+------------+
| Rohan     | Kumar    | Delhi | 9998712300 |
+-----------+----------+-------+------------+
1 row in set (0.00 sec)
```

03. View Score History

```sql
SELECT OldScore, NewScore, Reason, ChangeDate
FROM score_history
WHERE CitizenID = 101
ORDER BY ChangeDate DESC;
```

```
mysql> SELECT OldScore, NewScore, Reason, ChangeDate
    -> FROM score_history
    -> WHERE CitizenID = 101
    -> ORDER BY ChangeDate DESC;
+----------+----------+-------------+------------+
| OldScore | NewScore | Reason      | ChangeDate |
+----------+----------+-------------+------------+
|       70 |       85 | Tax payment | 2025-01-01 |
+----------+----------+-------------+------------+
1 row in set (0.00 sec)
```

04. View Actions Affecting Score

```sql
SELECT a.ActionName, a.ScoreImpact, r.ActionDate
FROM citizen_action_record r
JOIN action_type a
ON r.ActionTypeID = a.ActionTypeID
WHERE r.CitizenID = 101;
```

```
mysql> SELECT a.ActionName, a.ScoreImpact, r.ActionDate
    -> FROM citizen_action_record r
    -> JOIN action_type a
    -> ON r.ActionTypeID = a.ActionTypeID
    -> WHERE r.CitizenID = 101;
+-------------------+-------------+------------+
| ActionName        | ScoreImpact | ActionDate |
+-------------------+-------------+------------+
| Timely Tax Payment |          15 | 2025-01-01 |
| Illegal Parking    |          -5 | 2025-02-04 |
+-------------------+-------------+------------+
```

05. View Benefits Received

```sql
SELECT b.BenefitName, cb.GrantedDate
FROM citizen_benefit cb
JOIN benefit b
ON cb.BenefitID = b.BenefitID
WHERE cb.CitizenID = 101;
```

```
mysql> SELECT b.BenefitName, cb.GrantedDate
    -> FROM citizen_benefit cb
    -> JOIN benefit b
    -> ON cb.BenefitID = b.BenefitID
    -> WHERE cb.CitizenID = 101;
+--------------------+-------------+
| BenefitName        | GrantedDate |
+--------------------+-------------+
| Priority Loan      | 2025-02-01  |
| Fast Govt Service  | 2025-02-01  |
+--------------------+-------------+
2 rows in set (0.00 sec)
```

06. Check Eligible Benefits

```sql
SELECT BenefitName
FROM benefit
WHERE MinScoreRequired <=
(
SELECT CurrentScore
FROM citizen
WHERE CitizenID = 101
);
```

```
mysql> SELECT BenefitName
    -> FROM benefit
    -> WHERE MinScoreRequired <=
    -> (
    -> SELECT CurrentScore
    -> FROM citizen
    -> WHERE CitizenID = 101
    -> );
+-----------------------+
| BenefitName           |
+-----------------------+
| Priority Loan         |
| Fast Govt Service     |
| Tax Rebate            |
| Housing Priority      |
| Medical Assistance    |
| Education Scholarship |
+-----------------------+
6 rows in set (0.00 sec)
```

07. File Complaint

```sql
INSERT INTO complaint
(FiledByCitizenID, AgainstCitizenID, Description, Status, FiledDate)
VALUES (101,102,'Noise Pollution','Pending',CURDATE());
```

```
mysql> INSERT INTO complaint
    -> (FiledByCitizenID, AgainstCitizenID, Description, Status, FiledDate)
    -> VALUES (101,102,'Noise Pollution','Pending',CURDATE());
Query OK, 1 row affected (0.01 sec)
```

08. View Complaints Filed

```sql
SELECT ComplaintID, Description, Status
FROM complaint
WHERE FiledByCitizenID = 101;
```

```
mysql> SELECT ComplaintID, Description, Status
    -> FROM complaint
    -> WHERE FiledByCitizenID = 101;
+-------------+-----------------+----------+
| ComplaintID | Description     | Status   |
+-------------+-----------------+----------+
|           1 | Reckless driving | Reviewed |
|          23 | Noise Pollution  | Pending  |
+-------------+-----------------+----------+
2 rows in set (0.00 sec)
```

09. View Restrictions

```sql
SELECT RestrictionType, StartDate, EndDate
FROM restriction
WHERE CitizenID = 102;
```

```
mysql> SELECT RestrictionType, StartDate, EndDate
    -> FROM restriction
    -> WHERE CitizenID = 102;
+-----------------+------------+------------+
| RestrictionType | StartDate  | EndDate    |
+-----------------+------------+------------+
| Driving Ban     | 2025-01-20 | 2025-02-20 |
| Driving Ban     | 2026-03-11 | 2025-12-01 |
+-----------------+------------+------------+
2 rows in set (0.00 sec)
```

10. Compare Score with City Average

```sql
SELECT AVG(CurrentScore) AS CityAverage
FROM citizen
WHERE City =
(
SELECT City
FROM citizen
WHERE CitizenID = 101
);
```

```
mysql> SELECT AVG(CurrentScore) AS CityAverage
    -> FROM citizen
    -> WHERE City =
    -> (
    -> SELECT City
    -> FROM citizen
    -> WHERE CitizenID = 101
    -> );
+-------------+
| CityAverage |
+-------------+
|     83.2500 |
+-------------+
1 row in set (0.00 sec)
```

Government Official Dashboard Queries(10 Queries which an official dashboard can use)

01. View Pending Complaints

```sql
SELECT ComplaintID, FiledByCitizenID, AgainstCitizenID, Description
FROM complaint
WHERE Status = 'Pending';
```

```
mysql> SELECT ComplaintID, FiledByCitizenID, AgainstCitizenID, Description
    -> FROM complaint
    -> WHERE Status = 'Pending';
+-------------+------------------+------------------+-------------------+
| ComplaintID | FiledByCitizenID | AgainstCitizenID | Description       |
+-------------+------------------+------------------+-------------------+
|           3 |              104 |              105 | Littering         |
|          20 |              120 |              102 | Noise Disturbance |
|          22 |              110 |              108 | Illegal Parking   |
|          23 |              101 |              102 | Noise Pollution   |
|          24 |              101 |              102 | Noise Pollution   |
+-------------+------------------+------------------+-------------------+
5 rows in set (0.00 sec)
```

02. View Complaint Details with Citizen Information

```sql
SELECT c.ComplaintID,
f.FirstName AS FiledBy,
a.FirstName AS AgainstCitizen,
c.Description
FROM complaint c
JOIN citizen f ON c.FiledByCitizenID = f.CitizenID
JOIN citizen a ON c.AgainstCitizenID = a.CitizenID;
```

```
mysql> SELECT c.ComplaintID,
    -> f.FirstName AS FiledBy,
    -> a.FirstName AS AgainstCitizen,
    -> c.Description
    -> FROM complaint c
    -> JOIN citizen f ON c.FiledByCitizenID = f.CitizenID
    -> JOIN citizen a ON c.AgainstCitizenID = a.CitizenID;
+-------------+---------+----------------+-------------------+
| ComplaintID | FiledBy | AgainstCitizen | Description       |
+-------------+---------+----------------+-------------------+
|           1 | Rohan   | Aisha          | Reckless driving  |
|           2 | Suresh  | Neha           | Illegal parking   |
|           3 | Neha    | Kunal          | Littering         |
|           4 | Kunal   | Anita          | Water wastage     |
|           5 | Anita   | Rahul          | Traffic issue     |
|           6 | Rahul   | Pooja          | Noise pollution   |
|          20 | Aryan   | Aisha          | Noise Disturbance |
|          21 | Riya    | Kunal          | Public Littering  |
|          22 | Sunita  | Pooja          | Illegal Parking   |
|          23 | Rohan   | Aisha          | Noise Pollution   |
|          24 | Rohan   | Aisha          | Noise Pollution   |
+-------------+---------+----------------+-------------------+
11 rows in set (0.00 sec)
```

03. Record Citizen Action

```sql
INSERT INTO citizen_action_record
(CitizenID, ActionTypeID, OfficialID, ActionDate)
VALUES (102,1,1,CURDATE());
```

```
mysql> INSERT INTO citizen_action_record
    -> (CitizenID, ActionTypeID, OfficialID, ActionDate)
    -> VALUES (102,1,1,CURDATE());
Query OK, 1 row affected (0.01 sec)
```

04. Citizen with Violations

```sql
SELECT c.CitizenID, a.ActionName
FROM citizen_action_record r
JOIN action_type a ON r.ActionTypeID = a.ActionTypeID
JOIN citizen c ON r.CitizenID = c.CitizenID
WHERE a.ScoreImpact < 0;
```

```
mysql> SELECT c.CitizenID, a.ActionName
    -> FROM citizen_action_record r
    -> JOIN action_type a ON r.ActionTypeID = a.ActionTypeID
    -> JOIN citizen c ON r.CitizenID = c.CitizenID
    -> WHERE a.ScoreImpact < 0;
+-----------+------------------+
| CitizenID | ActionName       |
+-----------+------------------+
|       102 | Traffic Violation |
|       102 | Traffic Violation |
|       102 | Traffic Violation |
|       105 | Illegal Parking   |
|       109 | Illegal Parking   |
|       101 | Illegal Parking   |
|       107 | Littering         |
|       102 | Littering         |
+-----------+------------------+
8 rows in set (0.00 sec)
```

05. Citizen with Multiple Violations

```sql
SELECT CitizenID, COUNT(*) AS Violations
FROM citizen_action_record
GROUP BY CitizenID
HAVING COUNT(*) > 2;
```

```
mysql> SELECT CitizenID, COUNT(*) AS Violations
    -> FROM citizen_action_record
    -> GROUP BY CitizenID
    -> HAVING COUNT(*) > 2;
+-----------+------------+
| CitizenID | Violations |
+-----------+------------+
|       102 |          4 |
+-----------+------------+
1 row in set (0.00 sec)
```

06. Issue Restriction

```sql
INSERT INTO restriction
(CitizenID, DepartmentID, RestrictionType, StartDate, EndDate)
VALUES (102,1,'Driving Ban',CURDATE(),'2025-12-01');
```

```
mysql> INSERT INTO restriction
    -> (CitizenID, DepartmentID, RestrictionType, StartDate, EndDate)
    -> VALUES (102,1,'Driving Ban',CURDATE(),'2025-12-01');
Query OK, 1 row affected (0.04 sec)
```

07. View Restrictions by Department

```sql
SELECT r.RestrictionID, c.FirstName, r.RestrictionType
FROM restriction r
JOIN citizen c
ON r.CitizenID = c.CitizenID
WHERE r.DepartmentID = 1;
```

```
mysql> SELECT r.RestrictionID, c.FirstName, r.RestrictionType
    -> FROM restriction r
    -> JOIN citizen c
    -> ON r.CitizenID = c.CitizenID
    -> WHERE r.DepartmentID = 1;
+---------------+-----------+-----------------+
| RestrictionID | FirstName | RestrictionType |
+---------------+-----------+-----------------+
|             1 | Aisha     | Driving Ban     |
|             3 | Rahul     | Warning         |
|            21 | Arjun     | Driving Ban     |
|            22 | Aisha     | Driving Ban     |
|            23 | Aisha     | Driving Ban     |
+---------------+-----------+-----------------+
5 rows in set (0.00 sec)
```

08. Resolve Complaint

```sql
UPDATE complaint
SET Status = 'Reviewed'
WHERE ComplaintID = 3;
```

```
mysql> UPDATE complaint
    -> SET Status = 'Reviewed'
    -> WHERE ComplaintID = 3;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

09. Citizen with Complaints and Violations

```sql
SELECT DISTINCT c.CitizenID
FROM citizen c
JOIN complaint cm ON c.CitizenID = cm.AgainstCitizenID
JOIN citizen_action_record ar ON c.CitizenID = ar.CitizenID;
```

```
mysql> UPDATE complaint
    -> SET Status = 'Reviewed'
    -> WHERE ComplaintID = 3;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT DISTINCT c.CitizenID
    -> FROM citizen c
    -> JOIN complaint cm ON c.CitizenID = cm.AgainstCitizenID
    -> JOIN citizen_action_record ar ON c.CitizenID = ar.CitizenID;
+-----------+
| CitizenID |
+-----------+
|       102 |
|       104 |
|       105 |
|       106 |
|       107 |
|       108 |
+-----------+
6 rows in set (0.00 sec)
```

10. Most Frequent Violations

```sql
SELECT a.ActionName, COUNT(*) AS Frequency
FROM citizen_action_record r
JOIN action_type a
ON r.ActionTypeID = a.ActionTypeID
```

```sql
WHERE a.ScoreImpact < 0
GROUP BY a.ActionName
ORDER BY Frequency DESC;
```

```
mysql> SELECT a.ActionName, COUNT(*) AS Frequency
    -> FROM citizen_action_record r
    -> JOIN action_type a
    -> ON r.ActionTypeID = a.ActionTypeID
    -> WHERE a.ScoreImpact < 0
    -> GROUP BY a.ActionName
    -> ORDER BY Frequency DESC;
+------------------+-----------+
| ActionName       | Frequency |
+------------------+-----------+
| Traffic Violation |         3 |
| Illegal Parking   |         3 |
| Littering         |         2 |
+------------------+-----------+
3 rows in set (0.00 sec)
```

Analyst Dashboard Queries(10 Queries which an analyst dashboard can use)

01. Average Score By City

```sql
SELECT City, AVG(CurrentScore)
FROM citizen
GROUP BY City;
```

```
mysql> SELECT City, AVG(CurrentScore)
    -> FROM citizen
    -> GROUP BY City;
+------------+--------------------+
| City       | AVG(CurrentScore) |
+------------+--------------------+
| Ahmedabad |            55.0000 |
| Bhopal    |            91.0000 |
| Delhi     |            83.2500 |
| Indore    |            76.0000 |
| Jaipur    |            68.0000 |
| Kochi     |            82.0000 |
| Mumbai    |            55.0000 |
| Noida     |            60.0000 |
| Pune      |            69.0000 |
| Surat     |            72.0000 |
+------------+--------------------+
10 rows in set (0.00 sec)
```

02. Average Score By Age Group

```sql
SELECT AgeGroupID, AVG(CurrentScore)
FROM citizen
GROUP BY AgeGroupID;
```

```
mysql> SELECT AgeGroupID, AVG(CurrentScore)
    -> FROM citizen
    -> GROUP BY AgeGroupID;
+------------+--------------------+
| AgeGroupID | AVG(CurrentScore) |
+------------+--------------------+
|          1 |            65.0000 |
|          2 |            70.9000 |
|          3 |            90.5000 |
+------------+--------------------+
3 rows in set (0.00 sec)
```

03. Top Citizens

```sql
SELECT CitizenID, CurrentScore
FROM citizen
ORDER BY CurrentScore DESC
LIMIT 10;
```

```
mysql> SELECT CitizenID, CurrentScore
    -> FROM citizen
    -> ORDER BY CurrentScore DESC
    -> LIMIT 10;
+-----------+--------------+
| CitizenID | CurrentScore |
+-----------+--------------+
|       110 |           91 |
|       103 |           90 |
|       106 |           88 |
|       101 |           85 |
|       108 |           82 |
|       104 |           78 |
|       112 |           76 |
|       109 |           72 |
|       120 |           70 |
|       111 |           68 |
+-----------+--------------+
10 rows in set (0.00 sec)
```

04. Most Common Actions

```sql
SELECT ActionTypeID, COUNT(*)
FROM citizen_action_record
GROUP BY ActionTypeID
ORDER BY COUNT(*) DESC;
```

```
mysql> SELECT ActionTypeID, COUNT(*)
    -> FROM citizen_action_record
    -> GROUP BY ActionTypeID
    -> ORDER BY COUNT(*) DESC;
+--------------+----------+
| ActionTypeID | COUNT(*) |
+--------------+----------+
|            1 |        3 |
|            3 |        3 |
|            5 |        3 |
|            4 |        2 |
|            6 |        2 |
|            7 |        2 |
|            2 |        1 |
|            8 |        1 |
+--------------+----------+
8 rows in set (0.00 sec)
```

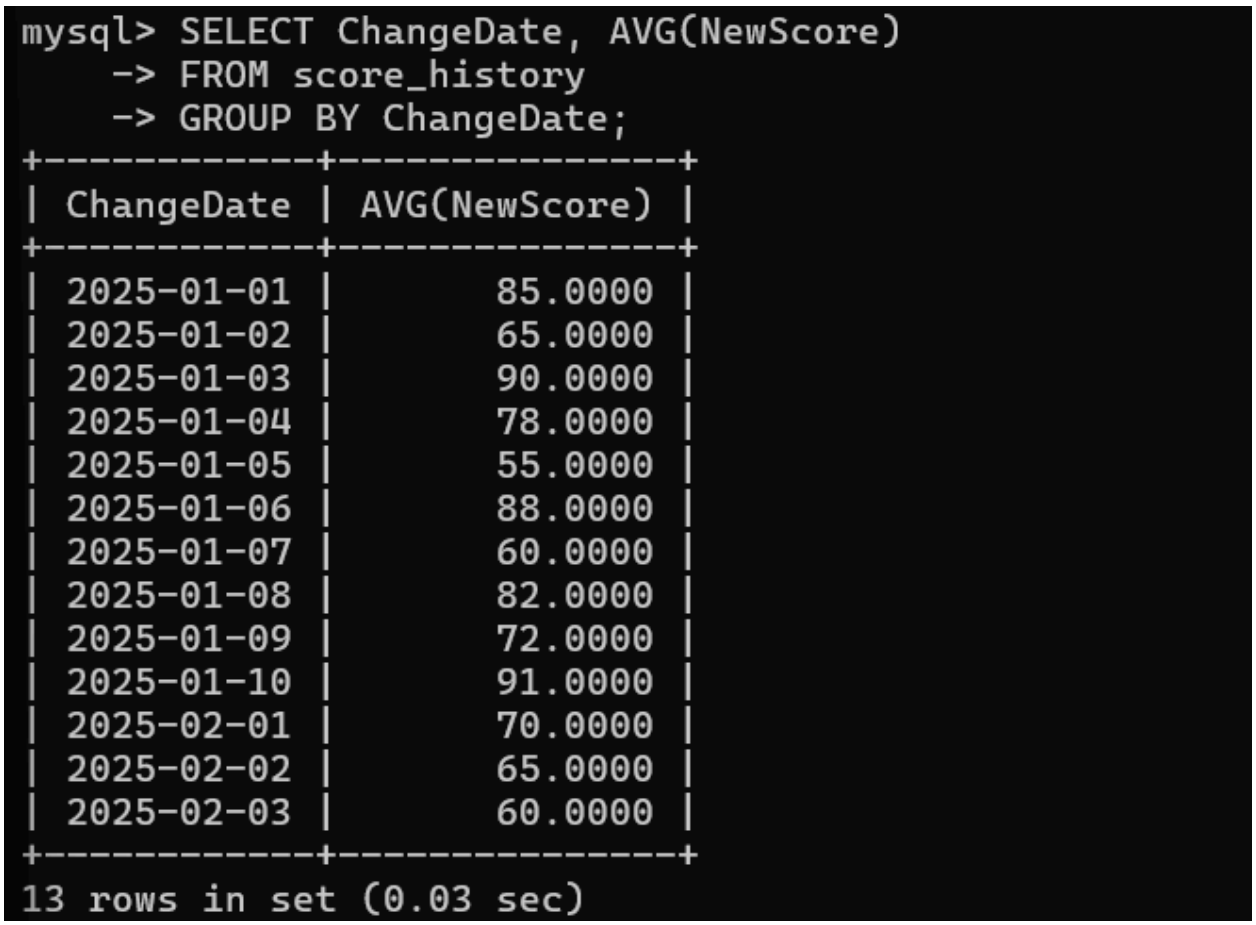
05. Complaint Distribution by City

```sql
SELECT c.City, COUNT(*) AS Complaints
FROM complaint cm
JOIN citizen c
ON cm.AgainstCitizenID = c.CitizenID
GROUP BY c.City;
```

```
mysql> SELECT c.City, COUNT(*) AS Complaints
    -> FROM complaint cm
    -> JOIN citizen c
    -> ON cm.AgainstCitizenID = c.CitizenID
    -> GROUP BY c.City;
+-----------+------------+
| City      | Complaints |
+-----------+------------+
| Mumbai    |          4 |
| Pune      |          1 |
| Ahmedabad |          2 |
| Delhi     |          1 |
| Noida     |          1 |
| Kochi     |          2 |
+-----------+------------+
6 rows in set (0.00 sec)
```

06. Score Trend Analysis

```sql
SELECT ChangeDate, AVG(NewScore)
FROM score_history
GROUP BY ChangeDate;
```



```
mysql> SELECT ChangeDate, AVG(NewScore)
    -> FROM score_history
    -> GROUP BY ChangeDate;
+------------+---------------+
| ChangeDate | AVG(NewScore) |
+------------+---------------+
| 2025-01-01 |       85.0000 |
| 2025-01-02 |       65.0000 |
| 2025-01-03 |       90.0000 |
| 2025-01-04 |       78.0000 |
| 2025-01-05 |       55.0000 |
| 2025-01-06 |       88.0000 |
| 2025-01-07 |       60.0000 |
| 2025-01-08 |       82.0000 |
| 2025-01-09 |       72.0000 |
| 2025-01-10 |       91.0000 |
| 2025-02-01 |       70.0000 |
| 2025-02-02 |       65.0000 |
| 2025-02-03 |       60.0000 |
+------------+---------------+
13 rows in set (0.03 sec)
```

07. Citizen with Highest Violations

```sql
SELECT CitizenID, COUNT(*) AS Violations
FROM citizen_action_record
GROUP BY CitizenID
ORDER BY Violations DESC;
```

```
mysql> SELECT CitizenID, COUNT(*) AS Violations
    -> FROM citizen_action_record
    -> GROUP BY CitizenID
    -> ORDER BY Violations DESC;
+-----------+------------+
| CitizenID | Violations |
+-----------+------------+
|       102 |          4 |
|       101 |          2 |
|       103 |          1 |
|       104 |          1 |
|       105 |          1 |
|       106 |          1 |
|       107 |          1 |
|       108 |          1 |
|       109 |          1 |
|       110 |          1 |
|       120 |          1 |
|       121 |          1 |
|       122 |          1 |
+-----------+------------+
13 rows in set (0.00 sec)
```

08. Citizen with Restrictions

```sql
SELECT c.CitizenID, r.RestrictionType
FROM restriction r
JOIN citizen c
ON r.CitizenID = c.CitizenID;
```

```
mysql> SELECT c.CitizenID, r.RestrictionType
    -> FROM restriction r
    -> JOIN citizen c
    -> ON r.CitizenID = c.CitizenID;
+-----------+-----------------+
| CitizenID | RestrictionType |
+-----------+-----------------+
|       102 | Driving Ban     |
|       105 | Fine Imposed    |
|       107 | Warning         |
|       109 | Tax Notice      |
|       108 | Fine Issued     |
|       111 | Driving Ban     |
|       102 | Driving Ban     |
|       102 | Driving Ban     |
+-----------+-----------------+
8 rows in set (0.00 sec)
```

09. Benefit Popularity

```sql
SELECT b.BenefitName, COUNT(cb.CitizenID)
FROM benefit b
LEFT JOIN citizen_benefit cb
ON b.BenefitID = cb.BenefitID
GROUP BY b.BenefitName;
```

```
mysql> SELECT b.BenefitName, COUNT(cb.CitizenID)
    -> FROM benefit b
    -> LEFT JOIN citizen_benefit cb
    -> ON b.BenefitID = cb.BenefitID
    -> GROUP BY b.BenefitName;
+----------------------+---------------------+
| BenefitName          | COUNT(cb.CitizenID) |
+----------------------+---------------------+
| Priority Loan        |                   2 |
| Fast Govt Service    |                   3 |
| Public Recognition   |                   2 |
| Tax Rebate           |                   0 |
| Housing Priority     |                   2 |
| Medical Assistance   |                   1 |
| Education Scholarship |                   1 |
+----------------------+---------------------+
7 rows in set (0.03 sec)
```

10. Citizens Eligible for All Benefits

```sql
SELECT CitizenID
FROM citizen
WHERE CurrentScore >=
(
SELECT MAX(MinScoreRequired)
FROM benefit
);
```

```
mysql> SELECT CitizenID
    -> FROM citizen
    -> WHERE CurrentScore >=
    -> (
    -> SELECT MAX(MinScoreRequired)
    -> FROM benefit
    -> );
+-----------+
| CitizenID |
+-----------+
|       103 |
|       110 |
+-----------+
2 rows in set (0.00 sec)
```